

Superstore Sales Analysis Report

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Dataset: superstore_dataset2011-2015.csv

Tools: Python (pandas, matplotlib), Jupyter Notebook

1. Introduction

This report looks at the Superstore sales data from 2011 to 2015 and tests five hypotheses about profit, sales, shipping and the timing of orders. The analysis was done in a Jupyter notebook with pandas and matplotlib. For each hypothesis the data was grouped and compared, and the results are shown below with a chart.

2. Data Overview

The dataset has 51,290 rows and 24 columns, and there are no duplicate rows. The only column with missing values is Postal Code, where 41,296 of the 51,290 rows are empty. I dropped it together with Row ID, Order ID and Customer ID, which left 20 columns.

From the summary statistics, the average sale is about 246.49, but the median is only 85.05 and the largest sale is 22,638.48, so a small number of very large orders pull the average up. Profit averages about 28.61 per row and goes from a loss of 6,599.98 to a gain of 8,399.98. The average discount is about 14.3% (maximum 85%), the average quantity is about 3.5 items per row, and the average shipping cost is about 26.38. There are four shipping modes: Standard Class, Second Class, First Class and Same Day.

3. Hypothesis Analysis

Hypothesis 1: Technology products have the highest profit margin

To test this I grouped the data by category and added up the profit, using total profit as the measure.

Category	Total Profit
Technology	~660,000
Office Supplies	~520,000
Furniture	~285,000

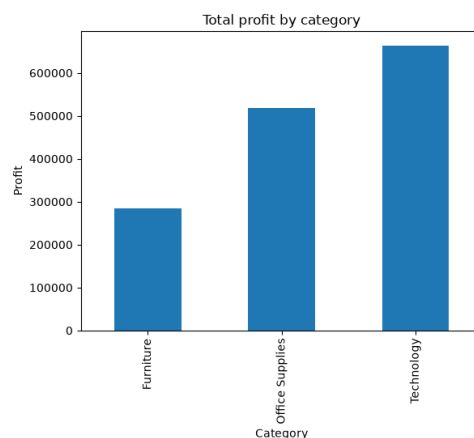


Figure 1: Total profit by category

Technology had the highest total profit, roughly 2.3 times the profit from Furniture. The hypothesis is supported.

Hypothesis 2: The East region has the highest sales

Here I added up sales for each of the 13 regions. Central was far ahead of the rest at about 2.8 million, followed by South at about 1.6 million and North at about 1.25 million. East came to only about 0.68 million, which puts it in the lower half, and Canada was the lowest at under 0.1 million.

Since Central has the highest sales and not East, this hypothesis is not supported.

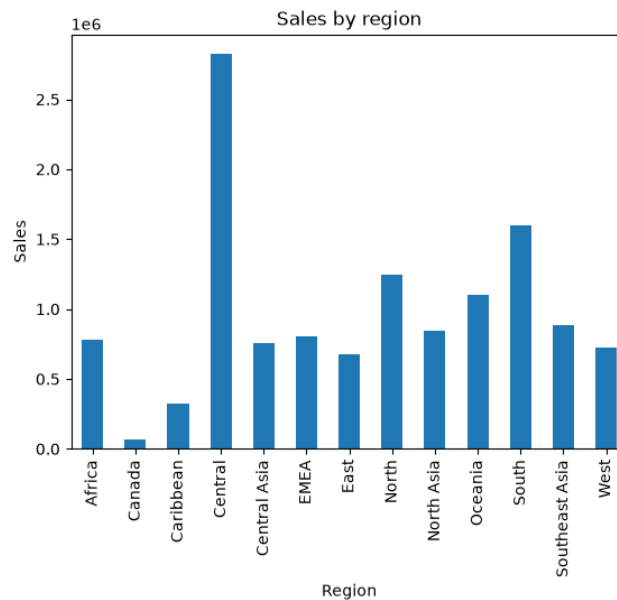


Figure 2: Total sales by region

Hypothesis 3: Sales are higher during certain months of the year

For this one I took the month from the order date and added up sales for each month across all the years. Sales were lowest in February at about 0.72 million and highest in November and December at about 1.38 million each, which is close to double February. Sales generally rise through the year, with dips in April, July and October.

This suggests that sales tend to increase toward the end of the year, so the hypothesis is supported.



Figure 3: Total sales by month

Hypothesis 4: Orders with same day shipping have the lowest rate of returned products

The dataset does not have a returns column, so I used orders with negative profit (loss-making orders) instead. For each shipping mode I calculated what percentage of its orders made a loss.

Ship Mode	Loss-making orders
Same Day	23.9%
Second Class	24.1%
Standard Class	24.5%
First Class	25.1%

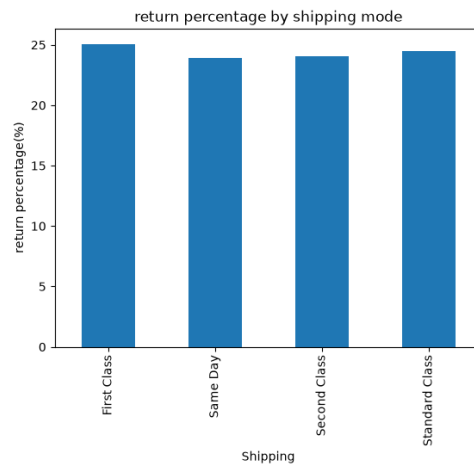


Figure 4: Loss-making orders by ship mode

Same Day had the lowest percentage of loss-making orders at 23.9%. However, the difference between the shipping modes is small (about 1.2 percentage points from lowest to highest), so shipping mode does not appear to have a major effect on this measure. The hypothesis is supported, but only by a small difference.

Hypothesis 5: The company's profit is more on weekdays than weekends

I got the day of the week from the order date and added up profit for each day.

Day	Total Profit
Monday	~246,000
Tuesday	~250,000
Wednesday	~205,000
Thursday	~241,000
Friday	~243,000
Saturday	~177,000
Sunday	~104,000

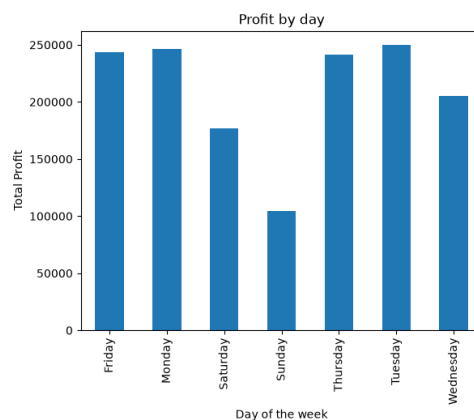


Figure 5: Total profit by day of the week

Looking at the data, the weekdays average about 237,000 profit per day compared with about 140,000 for Saturday and Sunday. Sunday is the lowest day, at less than half of Tuesday. The weekend gap holds even after accounting for there being five weekdays and only two weekend days, so the hypothesis is supported.

4. Key Findings

Technology has the highest total profit and Furniture the lowest.

Central is the top region for sales, well ahead of East, which was the region in the hypothesis.

Sales are lowest in February and highest in November and December.

About 24-25% of orders lose money in every shipping mode, and Same Day is only slightly lower than the others.

Profit on weekdays is clearly higher than on weekends, and Sunday is the weakest day.

5. Conclusion

Four of the five hypotheses were supported by the data, with the shipping mode result being only a small difference, and the East region hypothesis was not supported because Central had the highest sales. Overall the dataset shows a strong end-of-year sales pattern and higher profit on weekdays than on weekends.